

Vermont Health Platform: Research Lab & Tools

Comprehensive Reference Guide (2026 Edition)

1. Executive Summary: The Intelligence Layer

The Vermont Health Platform is architected as the cross-disciplinary intelligence layer between raw healthcare data and strategic decision-making. Operating on a **Six-Pillar Framework**, the platform provides a unified environment where Policy, Economics, Technology, Clinical, Equity, and Operations are analyzed as interdependent forces.

The **Research Lab** is the platform’s high-fidelity simulation engine. It allows users to transition from "Intelligence" (passive reading of articles) to "Execution" (active simulation and modeling).

2. The Six-Pillar Framework

2.1 Policy Pillar: "Is it Permissible?"

Focuses on the regulatory gates and legislative trajectories.

- **Core Intelligence:** Regulation & Legislation, Public Health Mandates, Global & Comparative Policy, Feasibility Studies.
- **The Gatekeeper Effect:** Policy analysis is the first interrogation point; an intervention that is clinically sound but regulatory prohibited is non-viable.

2.2 Economics Pillar: "Is it Sustainable?"

Analyzes financial viability and capital flows.

- **Core Intelligence:** Value-Based Care Models, Market & Finance, Labor Economics, Healthcare Investment Trends.
- **The Solvency Lens:** Tracks the "Scissors Effect" of rising fixed costs and falling FFS revenue in rural settings.

2.3 Technology Pillar: "Is it Possible?"

Evaluates the digital infrastructure required for scale.

- **Core Intelligence:** AI & Machine Learning, Digital Health, Data Security, Tech-Enabled Workflow.
- **Interoperability:** Focuses on FHIR R4 connectivity and technical deflation (reducing data exchange costs).

2.4 Clinical Pillar: "Is it Effective?"

Holds the system accountable to evidence-based outcomes.

- **Core Intelligence:** Hospital-at-Home, Precision Medicine, Virtual Care, Population Health.
- **Outcome Focus:** Metrics focus on preventable hospitalizations and readmission reduction.

2.5 Equity Pillar: "Is it Just?"

Measures the distributional impact of transformation.

- **Core Intelligence:** SDOH Integration, Algorithmic Bias, Access Disparity, Community Engagement.
- **Structural Focus:** Treats equity as an independent structural variable, not a derivative outcome.

2.6 Operations Pillar: "Is it Executable?"

Determines the organization's capacity to deliver.

- **Core Intelligence:** Revenue Cycle, Workforce Planning, Compliance & Risk, Supply Chain, Payer Network.
- **Execution Risk:** Identifies operational drag that prevents strategic intent from becoming reality.

3. The Research Lab (Pillar-Specific Tool Deep-Dive)

The Research Lab hosts 20 specialized tools designed for high-stakes modeling.

3.1 Policy Lab Tools

- **Policy Simulator:** A sandbox for modeling the legislative trajectory and regulatory feasibility of proposed health reforms.
- **Innovation Leaderboard:** A comparative index tracking the adoption and success rate of policy-driven health innovations across different jurisdictions.

3.2 Economics Lab Tools

- **APM Design Lab:** An actuarial workspace for designing Alternative Payment Models (APMs), including capitation and shared savings structures.
- **Shared Savings Calculator:** A tool for modeling the financial distribution between payers and providers under one-sided and two-sided risk tracks.
- **CEA Calculator:** A Cost-Effectiveness Analysis (CEA) tool utilizing QALY (Quality-Adjusted Life Year) benchmarks to determine the "value" of clinical interventions.
- **Hospital Financial Stress Test:** A solvency simulator that projects hospital margins and "days cash on hand" under various inflationary and volume scenarios.
- **HTA Studio:** A Health Technology Assessment module for evaluating the clinical and economic impact of new medical technologies.
- **Actuarial Lab:** A deep-dive environment for longitudinal cost-of-care modeling and population-level risk assessment.

3.3 Technology Lab Tools

- **FHIR Interoperability Lab:** A technical validation environment for testing data exchange standards and bidirectional HIE connectivity.
- **AI Clinical Governance Lab:** A structured framework for auditing AI safety, including the **Governance Builder** which assesses readiness across validation, transparency, and oversight.
- **Digital Health Lab:** A module focused on evaluating the ROI and operational reliability of virtual care and remote patient monitoring (RPM) platforms.

3.4 Clinical Lab Tools

- **Risk Stratification Engine:** A comprehensive clinical modeling tool featuring:
 - **HCC v28 Calculator:** Calculates CMS Risk Adjustment Factor (RAF) scores.
 - **Comorbidity Clustering:** Visualizes pairwise co-occurrence using Elixhauser and Charlson Index logic.

- **Custom Risk Builder:** Allows clinicians to build weighted composite scores for specific populations.
- **Clinical Quality Optimizer:** A dashboard for identifying gaps in evidence-based care bundles (e.g., Sepsis SEP-1, Maternal Hemorrhage).
- **Workforce Modeler:** A labor-economics tool for simulating the impact of staffing ratios and retention strategies on clinical throughput.

3.5 Equity Lab Tools

- **Population Health Modeler:** A geographic and demographic tool for mapping chronic disease prevalence against social determinants of health (SDOH).
- **Health Equity Studio:** Featuring the **Algorithmic Bias Detector**, which audits ML models for "Demographic Parity," "Equal Opportunity," and "Predictive Parity" across race, gender, and age.

3.6 Operations Lab Tools

- **Transformation Scorecard:** An executive-level progress tracker for organizational shifts from volume to value.
- **VBC Readiness Assessment:** A structured diagnostic for determining an organization's maturity across the five infrastructure layers required for Value-Based Care.
- **Research Workspace & Evidence Library:** A centralized repository for managing active simulation data and citing peer-reviewed evidence.

4. Global Platform Simulators & Signal Monitors

These 10 assets operate across the entire platform ecosystem.

4.1 Diagnostic Simulators

- **HTI Dashboard (Health Transformation Index):** The platform's flagship "Engine" quantifying maturity across six domains:
 1. **Digital Maturity** (20% weight)
 2. **Value-Based Care** (15% weight)
 3. **Social Determinants** (20% weight)
 4. **Clinical Excellence** (20% weight)
 5. **Patient Experience** (15% weight)
 6. **Workforce Wellness** (10% weight)
- **HTR Simulator:** A macro-level model for simulating the 10-year impact of "Technical Deflation" on state-wide GDP and total health expenditure.
- **Impact Simulation:** A high-level scenario modeler for predicting the "RPM Dividend"—the avoided cost of acute events through proactive monitoring.
- **Friction Index:** A metric measuring the "drag" created by administrative complexity, payer-provider misalignment, and clinical workflow inefficiencies.

4.2 Signal & Data Monitors

- **The Wire:** Live industry news feed filtered through the six pillars.
- **Investment Tracker:** Live monitor of venture capital flows, M&A activity, and capital deployment in health tech.
- **Trending Topics:** Real-time pulse of high-velocity issues like GLP-1 costs or PBM transparency.
- **Medicaid Eligibility Simulator:** A caseworker-grade tool for 5-step screening of program eligibility.
- **Multimedia:** Integrated podcast network and video briefings.
- **Six-Pillar Map:** The architectural visualizer for the HTR framework.

5. Core Methodologies & Technical Definitions

5.1 AI Performance Metrics

- **AUC-ROC:** Area Under the Receiver Operating Characteristic curve. Measures a model’s ability to distinguish between classes (e.g., patient likely to be readmitted vs. not).
- **F1 Score:** The harmonic mean of precision and recall. Essential for "imbalanced" healthcare data where true positives are rare.
- **NMV (Net Monetary Value):** Proprietary metric calculating: (Total Savings from True Positives) – (Cost of Intervention + Cost of False Positives).

5.2 Fairness & Equity Criteria

- **Demographic Parity:** Ensures the probability of a positive outcome is the same across groups (e.g., race, gender).
- **Equal Opportunity:** Ensures the True Positive Rate (Sensitivity) is identical across groups.
- **Predictive Parity:** Ensures the Positive Predictive Value (Precision) is identical across groups.

5.3 Clinical Risk Adjustments

- **HCC v28 (2024):** The latest CMS model focusing on chronic conditions and demographic variables.
- **Charlson Comorbidity Index:** Validated methodology for predicting 10-year survival.
- **Elixhauser Index:** Advanced methodology for predicting 30-day readmission risk based on 30 specific comorbidities.

6. User Persona Mapping

Persona	Key Lab Tools	Strategic Objective
Health System CFO	Stress Test, Shared Savings Calc	Protect operating margins during VBC transition.
Chief Medical Officer	Quality Optimizer, Risk Engine	Reduce readmissions and standardize care bundles.
Chief Tech Officer	FHIR Lab, AI Governance	Enable data liquidity and secure AI deployment.
State Health Official	Policy Simulator, Pop Health Modeler	Design permissible reforms and close equity gaps.
Strategy Analyst	HTA Studio, Investment Tracker	Evaluate new tech ROI and track market shifts.
Executive Lead	Scorecard, HTI Dashboard	Monitor organizational transformation milestones.